

SEMESTER-III

UMT404: DATA STRUCTURE				
	L	T	P	Cr
	2	0	2	3.0
Course Objective: To become familiar with different types of data structures and their applications.				
Syllabus Analysing algorithms: Basics of algorithm and its analysis, Complexity classes, order arithmetic, Time and space trade-off in algorithms. Linear Data Structures: Arrays, Strings and string processing, Linked lists (Singly, Doubly, Circular), Abstract data types, their implementation and applications: Stacks (using Arrays and Linked-list), Queues (using Arrays and Linked-list), Searching and Sorting: Linear Search, Binary Search. Introduction to internal and external sort, Bubble Sort, Selection Sort, Insertion Sort, Quick Sort, Merge Sort, Counting Sort. Trees and their applications: Introduction to binary tree, tree traversal algorithms, Binary search tree, AVL Tree, B Tree etc. and common operations on these trees. Heap, Heap Sort, Priority Queue using Heap. Graphs and their applications: Graph Terminology and its representation, Depth and breadth first traversals, Shortest-path algorithms (Dijkstra), Data Structures for Disjoint Sets, Minimum spanning tree (Prim and Kruskal).				
Laboratory Work Implementation of various data structures such as Arrays, Stacks, Queues, Lists, Binary tree traversals, BST, AVL trees, Graphs traversals, Sorting and Searching techniques.				
Course Learning Objectives (CLO) The students will be able to: 1. understand the fundamental data structures, their implementation and some of their standard applications. 2. select and implement appropriate searching and sorting techniques for solving a problem based on their characteristics. 3. apply tree and graph data structures for specific applications. 4. design and analyse algorithms using appropriate data structures for real-world problems.				
Text Books 1. Introduction to Algorithms, Cormen H. T., Leiserson E. C., Rivest L. R., and Stein C, MIT Press, 3rd ed., 2009 2. Data Structures, Algorithms and Applications in C++, Sahni S., Universities Press 2nd ed. 2005				

Approved in 113th meeting of the Senate held on September 7, 2024, respectively. Further, revised in 1st meeting of the Academic Council held on September 11, 2025.

Reference Books

1. Data Structures and Algorithms Made Easy, Karumanchi N., Career Monk Publications, 5th ed., 2017
2. Data structures and algorithms in C++, Adam Drozdek, 4th edition.

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1	MST	25
2	EST	40
3	Sessional (May include Assignments/Projects/Tutorials/Quizzes/Lab Evaluations)	35

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